Evaluating the Effects of a Nuclear Detonation in Low Earth Orbit

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ABSTRACT

High-altitude nuclear explosions pose risks to satellite constellations and ground-based infrastructure. We develop an open-source framework to evaluate first-order prompt radiation and delayed artificial radiation belts formed by beta-decay electrons from fission debris. The goal is to provide quantitative bounds on cascading space and terrestrial impacts of a nuclear detonation in low Earth orbit, where more than 13,000 satellites operate. The framework integrates x-ray transport, SGP4 orbital propagation, artificial radiation belt formation/evolution, electron total-ionizing-dose response, and electromagnetic pulse estimation. Using publicly available orbital data, we evaluate the impact of a nuclear detonation with a yield of 1 megaton TNT equivalent at burst altitude and subpoint optimized to maximize prompt satellite exposure. Prompt deposited x-ray fluence exceeding 40 J/m² disables approximately 4.4% and temporarily disables another 9.5% of the low Earth orbit population under these upper-bound geometries. The initial burst injects fission-beta electrons into the 2.15-2.40 L shell band quickly establishing a global and persistent artificial radiation belt 10^3-10^4 times greater than background. Within an hour after detonation, we find that about two-thirds of the satellites in low Earth orbit have already crossed the high-radiation region. Eventually, we estimate that ~98 percent of all satellites will receive disabling cumulative doses above a 10 krad(Si) threshold, within 30 days. In addition to effects in space, the EMP model indicates continental-scale ground-level impact.

INTRODUCTION

High-altitude nuclear detonations (HANEs) generate three coupled effects: prompt x-rays, artificial radiation belts, and electromagnetic pulse (EMP). The first example of these effects was demonstrated in the 1962 Starfish Prime test, a 1.4 MT detonation at 400 km, which created an artificial radiation belt that persisted for months disabling a significant fraction of the small satellite population then in orbit, and EMP induced blackouts in Hawaii (about 1,450 km away). Recent concerns about the possible deployment of a nuclear weapon in space, in violation of the Outer Space Treaty, have renewed interest in understanding the effects of a nuclear detonation with a focus on low Earth orbit, where key satellite constellations underpin global communications, navigation, Earth observation, weather forecasting, and national security operations.

As of January 2026, more than 13,000 objects operate in low Earth orbit (LEO, ≤2000 km) [1]. Satellite populations are increasingly concentrated near 500–600 km altitude due to the rapid deployment of large commercial constellations, increasing potential collateral effects from disruptive orbital events. Prior analyses have examined collateral satellite damage from EMP [2], and recent studies have estimated prompt HANE effects on LEO satellites [3]. Advances in x-ray transport modeling [4] and diffusion-based radiation belt frameworks such as DREAM [5] provide the relevant physical foundation. However, these mechanisms have not been evaluated publicly under present-day orbital densities within a unified, open modeling framework. Here, we couple radiation transport physics with contemporary satellite distributions to quantify prompt and secondary impacts.

METHODOLOGY

We develop *open-hane*, a first-order open framework comprising source term and burst region physics, magnetic field, orbit propagation, prompt x-ray transport, belt diffusion, and electron dose response. Each

module is checked against analytical limits, published reference values, and engineering approximation are explained.

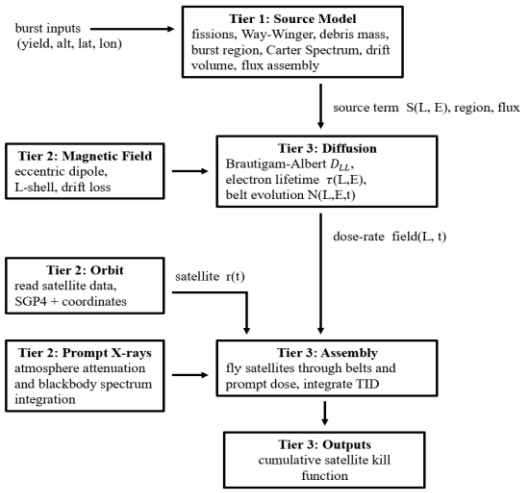


Figure 1. Data flow map of *open-hane* tiered code modules. Boxes represent modules that the code pipeline intakes and higher tiers are assigned when moving farther away from principal inputs.

The model requires burst parameters, radiation-source assumptions, magnetic-field approximation, and shielding assumptions. The test case presented here assumes a 1-megaton TNT equivalent (MT) burst at the peak concentration location: 485-km altitude, 40.75-degrees latitude, and -35.5-degrees longitude. Default parameters are presented in Table 1.

Table 1. Default Model Parameters. Parameters include yield, fission fraction, burst location, x-ray fraction, debris speed and kinetic fraction, atmospheric cutoff, and satellite shielding.

Parameter	Nominal value	Role in model
Yield	1 MT	Scales prompt x-ray energy and fission beta inventory.
Burst Altitude	485 km	Sets line-of-sight geometry and magnetic L-shell injection region.
Burst (lat, lon)	(+40.75, -35.5) deg	Sets burst location which defines burst region and trapping geometry.
Fission Fraction	0.5	Linearly scales beta-decay electron source [6].
X-ray Fraction	0.7	Scales prompt x-ray fluence [6].
Debris Speed	1500 km/s	Used to infer debris mass and burst expansion time [7].
Debris Kinetic Fraction	0.25	Sets kinetic energy in debris and magnetic-confinement radius [7].
Atmospheric Cutoff	100 km	Defines loss cone / drift-shell atmospheric loss boundary [10].
Shielding	2 mm Al	Equivalent shielding depth for dose response [2].

Orbital Population and Magnetic Field

General Perturbation element sets dated 14 January 2026 were propagated using SGP4 to a common epoch (2026-01-14T08:00:00Z) [1]. Of 14,117 cataloged objects, 13,306 satisfied the LEO criteria. Kernel density estimation shown in Figure 2 identified a concentration peak near 485 km altitude at latitude +40.75° and longitude −35.50°, defining an upper-bound burst geometry for this analysis.

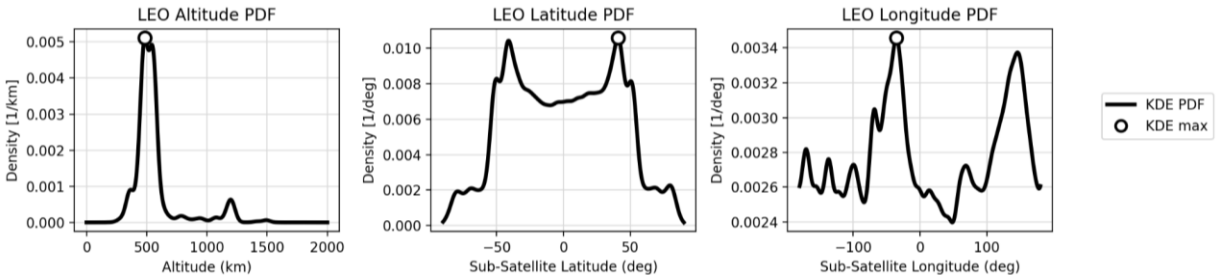


Figure 2. Kernel Density Estimation of LEO population density function in altitude, latitude and longitude. The densest population of satellites is estimated to be at latitude of +40.75 deg, longitude of -35.50 deg, and altitude of 485.0 km for epoch (2026-01-14T08:00:00Z) [1].

The geomagnetic geometry is modeled with an eccentric tilted dipole as a first-order approximation. The dipole axis is fixed to the geomagnetic pole (80.7° N, 72.7° W) and displaced from Earth's center by a magnitude of 450 km. For any position the model returns the dipole L-shell, magnetic latitude λ, and field strength:

$$B(L, \lambda) = (B_0 / L^3) \cdot \sqrt{(1 + 3\sin^2\lambda)} / \cos^6\lambda , \quad B_0 = 3.12 \times 10^{-5}\ \mathrm{T}$$

The field model is validated by setting the offset to zero to recover the analytic dipole, and the equatorial field minimum is recovered over the South Atlantic (reproducing the South Atlantic Anomaly). The accuracy of the magnetic dipole can be increased by using the IGRF field but costs more computationally.

Burst Region

The burst region is the spatial region the vaporized debris ions generated during a nuclear explosion immediately expands as a conducting plasma bubble. The extent of this region sets the L-shells that make up the injected belt. The debris mass (M_d) is derived from the debris kinetic energy (E_{KE}) where the kinetic energy is carried by the debris mass moving at speed U_d [7].

$$M_d = 2E_{KE}/(U_d{}^2)$$

The burst region extent in altitude (above and below burst) is determined by the smaller limit of the equal mass radius (R_m) and magnetic confinement radius (R_b). The model splits the debris equally in above/below burst hemispheres, with each assigned half the total debris mass and kinetic energy. The equal mass radius (R_m) is found when the debris expands an area of ambient mass equal to the debris mass in the hemisphere. The bubble will reach further upwards than downwards due to the change in atmospheric density. The magnetic confinement radius (R_b) is the distance when the geomagnetic field energy excluded by the bubble equals the kinetic energy; with magnetic energy density $B^2/2\mu_0$.

$$R_b = [\ 3\ (\tfrac{1}{2}\ E_{KE})\ /\ (\ 4\pi \cdot B^2/2\mu_0\ )\ ]^{1/3}$$

The radius extent in the above/below direction is the minimum of R_m and R_b (note the above is commonly defined by the magnetic radius while below is bound by equal mass radius). The radius limits are written as altitudes and converted to dipole L-shell bands. These bands define the initial trapping region of the artificial radiation belt. This was validated against results from Los Alamos National Laboratory Electron Source Model (ESM) (see Figure 3). It is noted that the magnetic confinement radius loses accuracy above 800km due to the simplified magnetic field model [7].

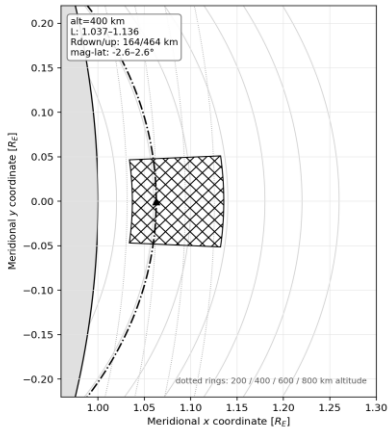


Figure 3. Burst radius for a 400km 1MT detonation along the equator. Simulation results showing the Earth (gray), burst location (black dot), and initial burst region (crosshatch). Results match very closely with the LANL Electron Source Model published unclassified burst region [7].

Prompt X-ray Transport

Prompt x-ray transport models the detonation as a blackbody emitter with energy scaled to yield (70% x-ray fraction). The vacuum fluence follows inverse-square scaling, and atmospheric attenuation is computed using energy-dependent mass attenuation coefficients with Planck spectral weighting [2,4], with line-of-sight occultation by the solid Earth. The blackbody spectrum is normalized and integrated using an upper bound of 50keV. The deposited x-ray fluence is calculated at each satellite position and compared against a 40 J/m² failure threshold and a 0.01 J/m² reversible-upset threshold [3]. The fluence Φequation is:

$$\Phi = \frac{Y\, f_x\, E_{kt}}{4\pi R^2} \cdot \frac{\int_{E_{min}}^{E_{max}} B(E,kT) e^{-\mu(E)\sigma} dE}{\int_{E_{min}}^{E_{max}} B(E,kT) dE}$$

where Y is the burst yield [kt], f_x is the x ray fraction of yield, $E_{kt} = 4.184E12\ [J/kt]$, R is the slant range from burst [m], the Planck blackbody spectrum at temperature kT is $B(E, kT) = (15/\pi^4)\,(kT)^{-4} \cdot E^3 / (e^{E/kT} - 1)$, $\mu(E)$ is the energy dependent mass attenuation coefficient of air [cm²/g] [4], σ is the slant atmospheric column of mass along the line of sight [g/cm²] and the energy integrations limits are 0.1 to 50 keV. The atmospheric model is a replica of the ESM ionosphere profile [7].

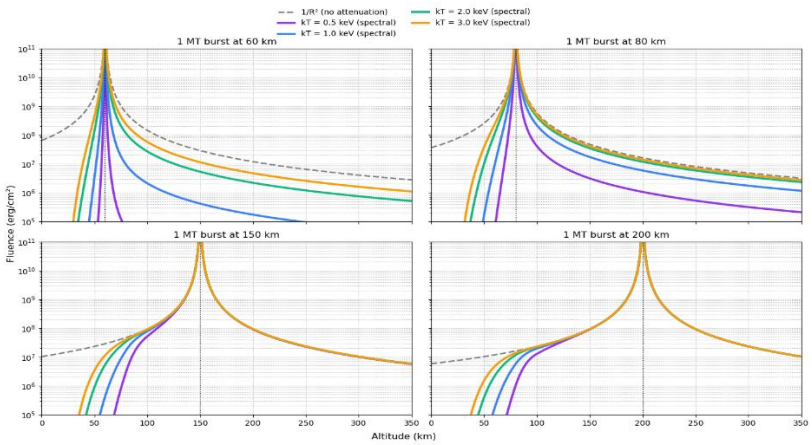


Figure 4. Attenuation of blackbody spectrum with temperatures ranging from 0.5 to 3 keV at altitudes spanning 60km to 200km. The model output is plotted against results from *Collateral Damage to Satellites from an EMP Attack* Figure V.6 with best matching at higher temperatures and altitudes [2].

Source Term and Artificial-Belt Injection

The number of fissions is determined by the yield and fission fraction with a 1.45 x 10^{23} fissions/kt conversion factor [3,6]. Trapping physics then determines what fraction of isotopically emitted electrons contribute to the artificial radiation belt or are absorbed into the atmosphere.

A trapped electron will bounce along a magnetic field line between mirror points and drift in longitude around the Earth's magnetic axis. Its magnetic moment (first adiabatic invariant) is conserved and $\sin^2\alpha$ / B is constant along the field line, where α is the local pitch angle [8]. The reversal of direction at the mirror point occurs when the parallel velocity is zero and all momentum is perpendicular to motion. At this location the magnetic field can be defined as $B_m = B_{eq}/sin^2\ \alpha_{eq}$, with α_{eq} the equatorial pitch angle and B_{eq} the equatorial field strength. An electron is lost when its mirror point lies below the atmospheric cut off altitude or equivalently if α_{eq} is less than the loss cone angle α_{LC} defined by $sin^2\ \alpha_{LC} = B_{eq}/B(h_{atm})$ [8,9,10]. The retained fraction of electrons per shell, $\eta(L)$, is the loss cone evaluated at the weakest field point along the full longitudinal drift path. The trapping efficacy approximation is an upper bound and is static. Approximations include no drift losses (from South Atlantic Anomaly), and no plasma-wave (pitch-angle) scattering [11].

The fission beta source has a temporal and spectral factor that is computed at each time step. The temporal factor follows the Way-Wigner decay law. The number of beta electrons emitted per fission between times t_1 and t_2 is obtained by integrating the decay rate.

$$\eta_\beta(t_1, t_2) = (A/0.2)\,(\,{t_1}^{-0.2} - {t_2}^{-0.2}\,),\ A = C_{WW}/\,\bar{E}_\beta$$

Where the fission-produce activity decays as $t^{-1.2}$, the Way-Wigner coefficient $C_{WW} = 1.26\ MeV \cdot s \cdot fissions^{-1}$, and $\bar{E}_\beta = 1.358$ MeV is the mean beta energy (source starts at one second after burst) [10,12]. The spectral factor follows the Carter fission-beta spectrum and is evaluated on an electron energy grid and parametrized analytically as

$$J(E) \propto \beta\, e^{-0.2938(\gamma-1)-0.0144(\gamma-1)^2}$$

with E = kinetic energy [MeV], $\gamma = 1 + \frac{E}{0.511}$ (Lorentz factor), and $\beta = \sqrt{1 - 1/\gamma^2}$ (speed units c) [13]. The spectrum is normalized to unit integral over the energy grid (50 keV to 8 Mev in 200 logarithmically spaced steps). At each time step the electrons injected are the product of the total fission number, Way-Wigner increment η_β over the interval, and the energy spectrum which the population is distributed with.

Belt Evolution and Dose Rate

The belt extents evolve on a grid from L= [1.05, 6] and energy range [50keV-8MeV]. At each time step the source injects electrons emitted in a specified interval with distributed energy by the Carter spectrum. The electrons are spread across the L-shells with a Gaussian centered on the burst shell (its width set by the injection band) and scaled by the trapping efficiency $\eta(L)$. The trapped population N(L,E,t) evolves under radial diffusion. The diffusion equation incorporates spreading of electrons in the L shells (term 1), losses via scattering/absorption/decay (term 2), and the source term, S.

$$\partial N/\partial t = \partial/\partial L\,(D_{LL}\,\partial N/\partial L) - N/\tau(E,L) + S(L,E,t)$$

The radial diffusion coefficient is the Brautigam–Albert parameterization [14], $D_{LL} = 10^{(0.506Kp-9.325)}L^{10}$ per day, evaluated at a moderate geomagnetic-activity index, Kp = 2. The differential equation is solved using backward-Euler (implicit) scheme solved by a tridiagonal algorithm under zero-flux boundaries. The electron lifetime term τ, is an approximation and is not derived (codes such as DREAM [5] produce rigorous values). Also, the transport uses the plain radial-diffusion operator rather than the full phase-space form $\partial/\partial L[L^2\,D_{LL}\partial/\partial L(N/L^2\,)]$. The belt exists with a characteristic 1/e decay time. The electron number density n=N/V(L) uses the trapped shell volume (proportional to $\int\cos^7\lambda\ d\lambda$) to determine electrons per m^3.

The trapping efficiency in the code is an upper bound approximation because of missing loss terms such as plasma-wave pitch-angle scattering. Therefore, the model separately reports a calibrated 10% trapping efficiency to mimic the results of LANL's Electron Source model for a 1MT burst at 400km over the Pacific [19]. The differences in results are minimal and are discussed later in the results section.

Flux is obtained by multiplying the number density by electron speed, $\phi = n \cdot v(E)$. Finally, the dose rate is determined by multiplying the flux by the dose an electron deposits in silicon behind a fixed 2 mm Al equivalent shield R(E) and sum over energy. The code produces a dose rate (DR) map as a function of position and time that integrates the rad/s satellites experience when their orbit is traced.

$$DR = \Sigma \phi(E) R(E)$$

NIST ESTAR data is used to compute the dose deposited behind the 2mm Al equivalent shield. The dose response per incident electron is:

$$R(E) = S_{Si(E)} T_{Al(E)} 1.602 \times 10^{-8} \ [rad(Si)\ per\ electron\ cm^{-2}]$$

where $S_{Si(E)}$ is the silicon collision stopping power [MeV·cm^2/g], the factor converts MeV/g per particle·cm^2 to rad, and $T_{Al(E)}$ is the shield transmission. The transmission is modeled as a range-based attenuation through the Al column using CSDA range from NIST ESTAR [15], with a sharpened cutoff once the shield thickness exceeds the electron range and includes a small bremsstrahlung term [13]. The dose is an approximation and can be improved by using software such as SHIELDOSE-2 which is done in modeling radiation environments for NASA applications [20].

The satellites are propagated and at each timestep the dipole L-shell is calculated, and the corresponding local belt dose rate is found. The dose rate is time-integrated to give the delayed belt total ionizing dose (TID). The total per-satellite dose is the sum of the prompt (occurs at time of burst) and the delayed contributions (start one second after burst). The satellite is considered disabled when its TID reaches 10 krad(Si) but is also compared at higher limits of 100 krad(Si) and 300 krad(Si) that represent hardened or military LEO satellites [2, 16, 17].

Electromagnetic pulse

The fast (E1) component of the high-altitude EMP is modeled using a modified Karzas–Latter–Seiler code available openly [18]. The results are used to show the geographic extent of the large ground impact and is independent of satellite effects.

Results

A 1MT burst at 485 km altitude, +40.75° latitude and −35.50° longitude creates a prompt x-ray fluence exceeding the 40 J/m² failure threshold killing 584 satellites (4.4% of the LEO population) with an additional 1,267 satellites (9.52 %) receiving fluence in the 0.01-40 J/m² reversible-upset threshold. The prompt kill fraction scales with yields as 0.53%, 4.39%, and 12.93% for 0.1, 1 and 10 MT respectively. Figure 5 shows prompt failures localized about the burst subpoint and have greater impact on higher altitude satellites due to less atmospheric attenuation.

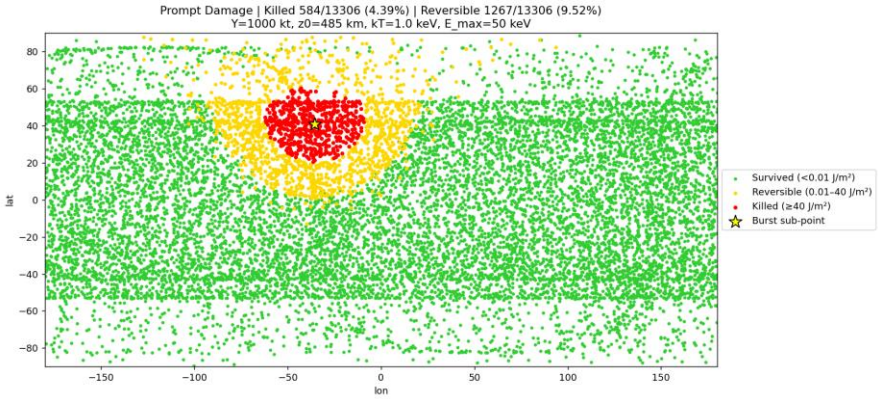


Figure 5. Prompt x-ray satellite damage from a 1MT burst at 485 km altitude, +40.75° latitude and −35.50° longitude. Green represents satellites that survived, yellow represents possible irreversible damage to the satellite, and red represents killed satellites. We find that 4.4% (584) of satellites are directly destroyed and 9.5% (1,267) experience possibly reversible damage.

The same high altitude nuclear explosion injects fission-beta electrons into geomagnetic field lines initially spanning L shells 2.15 to 2.40. The trapped electrons drift in longitude to populate a complete shell within minutes. The resulting belt is global in extent and threatens any satellite that crosses the injected L band, independent of burst longitude. For this detonation location the trapped electron shell descends to LEO altitudes at high magnetic latitudes such that the satellites will enter the belt near the poles on each orbit. Tracing the full catalog through the geometry showed 64% of LEO satellites crossing the injected band within the first hour and repeatedly after.

For every LEO satellite, the delayed-belt total ionizing dose (TID) is found by tracing the satellite through the evolving dose-rate field and integrating local dose rate over time, behind a 2 mm Al-equivalent shield. There are two quantities reported throughout: the geomagnetic upper bound computed with static trapping efficiency (approximate dipole and no wave scatter losses) and the LANL Electron Source Model (ESM) net trapping efficiency of 10% for a comparable geometry [19].

The delayed belt is the dominant kill mechanism. The prompt effect kills 4.4% of LEO satellites instantaneously, but the accumulated belt TID causes extreme satellite losses over the following weeks (Figure 6). Over a 30-day window the upper bound places 98.2% of LEO satellites above the 10 krad(Si) failure dose; the ESM-calibrated estimate yields 97.5%. The two estimates are comparable because the dose exceeds the 10 krad(Si) thresholds by orders of magnitude and therefore the trapping reduction will not have a large impact on the total satellites killed.

The cumulative-loss curve begins with the prompt fraction at time zero and rises as the artificial belt dose accumulates on the remaining satellites. The loss is rapid as shown in Figure 6 where cumulative satellite loss as a function of time is shown for multiple shielding options. Even a 300 krad(Si) hardened threshold, which is 6 to 30 times higher than commercial LEO satellites [20], with a 10% trapping efficiency will reach 90% failure within 62 days according to the model. Overall, we find that over 90% of all satellites are disabled within 30 to 60 days across the radiation thresholds and trapping efficiencies included in this analysis. The comparison of the results across the failure threshold range and trapping efficiency range is summarized in Figure 7.

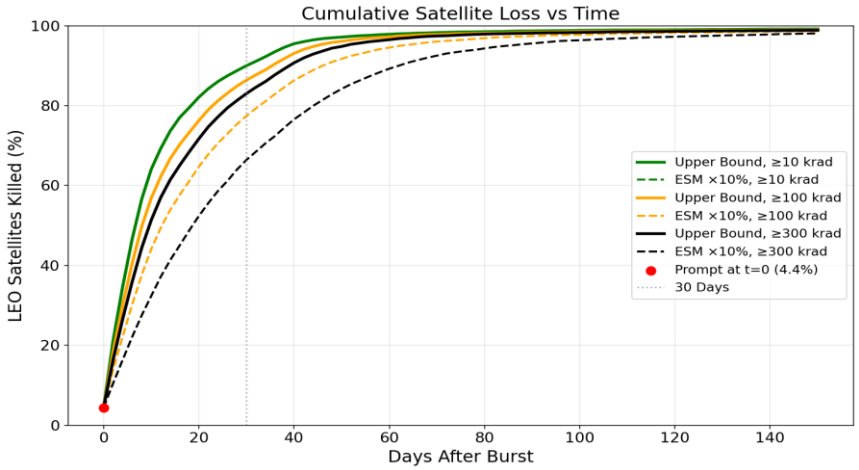


Figure 6. Cumulative satellite losses over time from a 1MT burst at 485 km altitude, +40.75° latitude and −35.50° longitude. Three different shielding thresholds are used representing commercial (10krad(Si)) and hardened military satellites (100krad(Si) and 300krad(Si)) and are compared against a calibrated 10% trapping efficiency results. Note that the 10% trapping efficiency curve for ≥ 10krad is equivalent to the upper bound curve for ≥ 100krad.

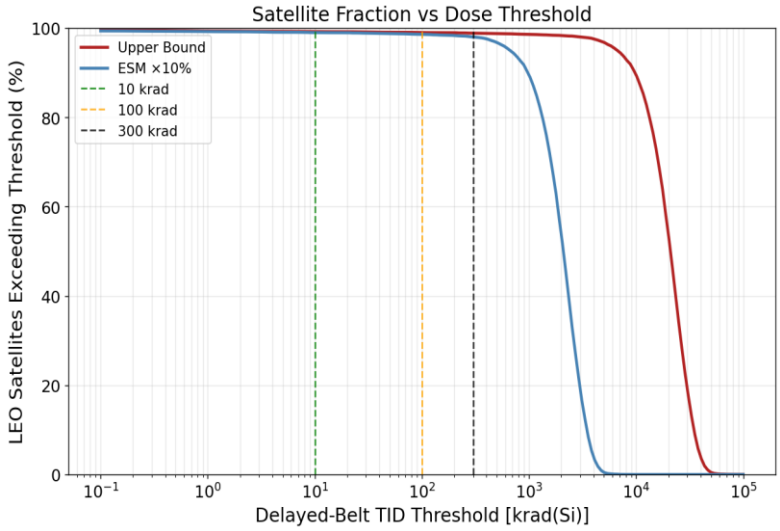


Figure 7. Percentage of LEO satellites above a specified total ionizing dose for two trapping efficiencies. The vertical dashed lines show various TID kill thresholds which do not have significant differences between the upper bound and 10% trapping efficiency. The TID is calculated for lifetime of the belt ~150 days.

To compare the artificial belt to natural background levels, the accumulated dose is compared to the dose a satellite would receive from the natural trapped-electron belt over the same window and behind the same shield. Using the LEO trapped-electron environment (integral flux ~$10^4/cm^2 \cdot s$ in benign regions to ~$10^6/cm^2 \cdot s$ near the poles and South Atlantic Anomaly), converted to dose through the same response function, the artificial-belt dose exceeds the natural belt by an approximate factor on the order of 10^3-10^4 [20]. This order of magnitude matches a claim by the Defense Threat Reduction Agency High-Altitude Nuclear Detonations Against Low Earth Orbit Satellites (HALEOS) study [21].

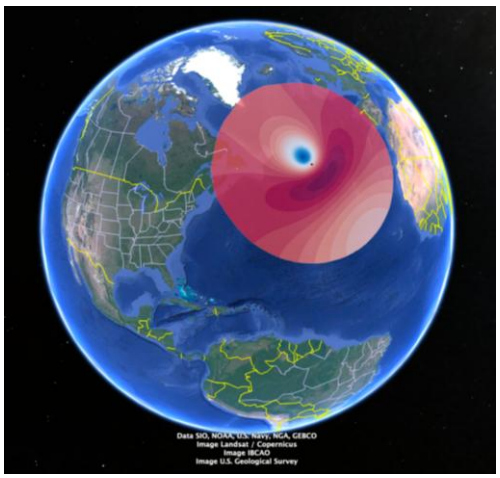


Figure 8. Globe plot of Karzas–Latter–Seiler formulation for the (E1) component of a 1MT burst at 485 km altitude, +40.75° latitude and −35.50° longitude. The colors match a logarithmic scale from dark blue (0V/m) to deep red (135,000V/m).

The Karzas–Latter–Seiler formulation of the (E1) component of a 1MT burst at 485 km altitude, +40.75° latitude and −35.50° longitude is shown in Figure 8. The results show an approximate circle with diameter ~4,400km and peak electric field of 135,000V/m. Because of the burst coordinates, the E1 electric field would primarily occur above the North Atlantic Ocean. The magnitude suggests, however, that a burst directly above land would lead to continental-scale ground damage to electrical equipment and grid.

Conclusion

An open-source code was developed to model a 1MT nuclear explosion at 485km to estimate prompt effects, delayed radiation, and EMP generation. The prompt x-rays killed about 4.4% of LEO satellites instantaneously in the local region around the burst. The delayed artificial radiation belt produced by the HANE is global and persistent causing 98% of LEO satellites to exceed a 10 krad(Si) threshold within 30 days. This result holds even against the 300 krad(Si) hardened threshold. The high-altitude electromagnetic pulse (E1) produces a peak ground level electric field of ~135,000V/m over a circular footprint roughly 4,400km in diameter, a continental threat to ground and airborne electronic infrastructure. Of the three effects the artificial radiation belt is the dominant threat to orbital constellations from its multi-month-long persistence and capacity to degrade most LEO satellites within weeks. These results together show that even a single high-altitude nuclear detonation would have indiscriminate, regional and global effects in space and on the ground including civilian infrastructure. Multiple bursts, including at locations meant to maximize trapping efficiency, would likely increase the magnitude and speed at which all LEO satellites would be disabled. It is fundamental to uphold the current Outer Space Treaty that bans the deployment of nuclear weapons in orbit.

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